

Midterm Exam

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1 Artificial Intelligence and Application Midterm Exam

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1.1 Task 1

To load and process data I will use pandas library and for drawing graph I will use pyplot from matplotlib.

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
```

Firstly, let's load data and convert datetime strings to datetime.

```
[2]: data_frame = pd.read_csv('data/2024-03-transaction.csv')

data_frame['timestamp'] = pd.to_datetime(data_frame['timestamp'])
```

Next, let's filter data from 2024-03-11 to 2024-03-19 and extract hour from each timestamp.

```
[3]: start_date = '2024-03-11'
end_date = '2024-03-19'
data_frame = data_frame[(data_frame['timestamp'] >= start_date) &
    ↪(data_frame['timestamp'] < end_date)]

data_frame['hour'] = data_frame['timestamp'].dt.floor('h')
```

Finally, use groupby on hour and side to aggregate number of transactions during each hour.

```
[4]: transaction_counts = data_frame.groupby(['hour', 'side']).size().
    ↪unstack(fill_value=0)
transaction_counts.columns = ['buy', 'sell']
```

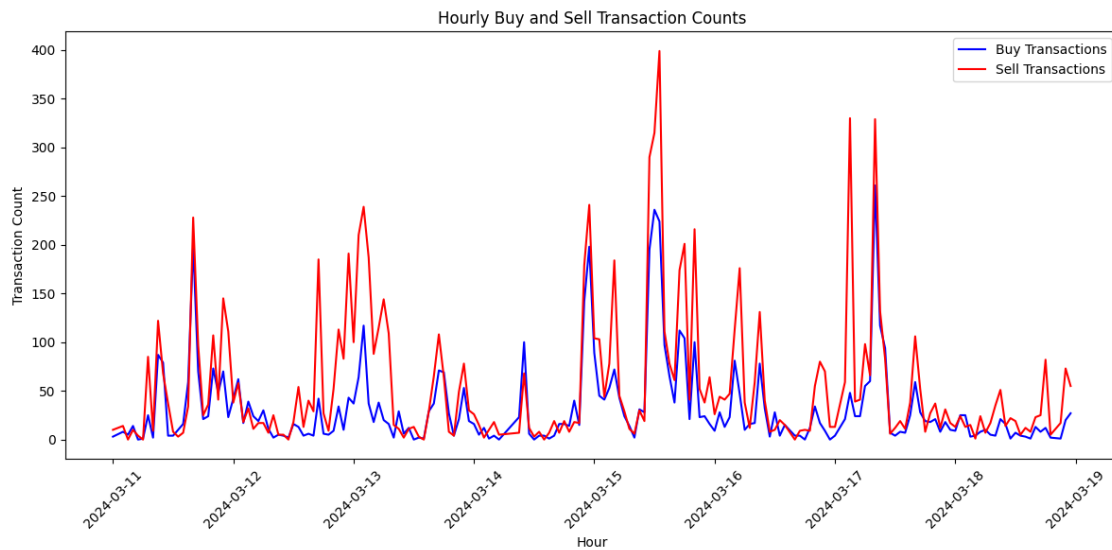
Now, let's create plot with output data.

```
[5]: plt.figure(figsize=(12, 6))
plt.plot(transaction_counts.index, transaction_counts['buy'], label='Buy',
    ↪Transactions', color='blue')
```

```
plt.plot(transaction_counts.index, transaction_counts['sell'], label='Sell_↵
↵Transactions', color='red')

plt.xlabel('Hour')
plt.ylabel('Transaction Count')
plt.title('Hourly Buy and Sell Transaction Counts')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```



1.2 Task 2

Firstly, data loading:

```
[6]: transactions = pd.read_csv('data/2024-03-11-transaction.csv')
raw_data = pd.read_csv('data/2024-03-11-rawdata.csv')
```

Secondly, let's parse datetime strings and remove us from raw_data timestamps.

```
[7]: transactions['timestamp'] = pd.to_datetime(transactions['timestamp'],↵
↵format='%Y-%m-%d %H:%M')
raw_data['timestamp'] = pd.to_datetime(raw_data['timestamp'], format='%Y-%m-%d↵
↵%H:%M:%S.%f').dt.floor('s')
```

Next, I will remove all raw_data records, that has second part of timestamp set to other value than 0. This is because transactions file has timestamps only in full minutes (which implies second set to 0).

```
[8]: raw_data = raw_data[raw_data['timestamp'].dt.second == 0]
```

Next, let's use `groupby` to group `raw_data` by timestamp.

```
[9]: grouped_raw_data = raw_data.groupby('timestamp')
```

Now, I can define function that will calculate new features based on single group.

```
[10]: def calculate_features(orderbook_slice):
    top_bid_price = orderbook_slice[orderbook_slice['type'] == 0]['price'].
    ↪iloc[0]
    top_ask_price = orderbook_slice[orderbook_slice['type'] == 1]['price'].
    ↪iloc[0]
    midprice = (top_ask_price + top_bid_price) * 0.5

    askQty = orderbook_slice[orderbook_slice['type'] == 1]['quantity'].max()
    bidQty = orderbook_slice[orderbook_slice['type'] == 0]['quantity'].max()
    bidPx = orderbook_slice[orderbook_slice['type'] == 0]['price'].mean()

    bookfeature = (askQty * bidPx) / bidQty
    alphaprice = bookfeature - midprice

    return pd.Series([midprice, bookfeature, alphaprice], index=['midprice',
    ↪'bookfeature', 'alphaprice'])
```

Finally, I will apply `calculate_features` function to each group.

After this I can merge `raw_data` with transactions, using keys from left table (transactions).

```
[ ]: features = grouped_raw_data.apply(calculate_features).reset_index()

transactions = pd.merge(transactions, features, left_on='timestamp',
    ↪right_on='timestamp', how='left')

transactions = transactions[['midprice', 'bookfeature', 'alphaprice',
    ↪'timestamp', 'quantity', 'price', 'side']]
```

The output:

```
[12]: result = pd.concat([transactions.head(10), transactions.tail(10)])
print(result)

output_file = 'data/2024-03-11-transaction-modified.csv'
transactions.to_csv(output_file, index=False)
```

	midprice	bookfeature	alphaprice	timestamp	quantity	\
0	97798500.0	2.628835e+07	-7.151015e+07	2024-03-11 00:07:00	0.150000	
1	97798500.0	2.628835e+07	-7.151015e+07	2024-03-11 00:07:00	0.010229	
2	97798500.0	2.628835e+07	-7.151015e+07	2024-03-11 00:07:00	0.001024	

3	97784500.0	4.044031e+07	-5.734419e+07	2024-03-11	00:11:00	0.052561
4	97597000.0	2.048226e+08	1.072256e+08	2024-03-11	00:17:00	0.039159
5	97597000.0	2.048226e+08	1.072256e+08	2024-03-11	00:17:00	0.070000
6	97597000.0	2.048226e+08	1.072256e+08	2024-03-11	00:17:00	0.037624
7	97597000.0	2.048226e+08	1.072256e+08	2024-03-11	00:17:00	0.002864
8	97597000.0	2.048226e+08	1.072256e+08	2024-03-11	00:17:00	0.001092
9	97849000.0	1.374677e+07	-8.410223e+07	2024-03-11	00:34:00	0.001248
2037	101286000.0	1.052174e+09	9.508882e+08	2024-03-11	23:31:00	0.004640
2038	101286000.0	1.052174e+09	9.508882e+08	2024-03-11	23:31:00	0.000099
2039	101286000.0	1.052174e+09	9.508882e+08	2024-03-11	23:31:00	0.001777
2040	101286000.0	1.052174e+09	9.508882e+08	2024-03-11	23:31:00	0.045300
2041	101046500.0	2.899559e+07	-7.205091e+07	2024-03-11	23:40:00	0.180088
2042	101046500.0	2.899559e+07	-7.205091e+07	2024-03-11	23:40:00	0.019912
2043	101233500.0	2.442022e+08	1.429687e+08	2024-03-11	23:51:00	0.002059
2044	101233500.0	2.442022e+08	1.429687e+08	2024-03-11	23:51:00	0.000494
2045	101233500.0	2.442022e+08	1.429687e+08	2024-03-11	23:51:00	0.003312
2046	101233500.0	2.442022e+08	1.429687e+08	2024-03-11	23:51:00	0.019000

	price	side
0	97789000	1
1	97727000	1
2	97700000	1
3	97694000	0
4	97704000	1
5	97704000	1
6	97704000	1
7	97718000	1
8	97709000	1
9	97773000	1
2037	101267000	1
2038	101267000	1
2039	101268000	1
2040	101240000	1
2041	101016000	0
2042	101016000	0
2043	101192000	1
2044	101192000	1
2045	101192000	1
2046	101192000	1

1.3 Task 3

To create a smart trading agent I can use `side` as a target feature, since we are operating on historical transactions data. This target will train the model to predict whether the agent should buy or sell based on the features derived from Task 2.

Firstly, import necessary libraries and load data.

```
[13]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
import seaborn as sns

data_frame = pd.read_csv('data/2024-03-11-transaction-modified.csv')
```

Next, let's choose features and target.

```
[14]: X = data_frame[['midprice', 'bookfeature', 'alphaprice']]
X = StandardScaler().fit_transform(X)
Y = data_frame['side']
```

Next, we can use `train_test_split` to split data into training and testing datasets.

```
[15]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3,
↳ random_state=101)
```

After splitting dataset, we can train the model using `RandomForestClassifier`.

```
[16]: model = RandomForestClassifier(n_estimators=700, random_state=101)
model.fit(X_train, Y_train)
```

```
[16]: RandomForestClassifier(n_estimators=700, random_state=101)
```

As model is trained I can use it to make some predictions and evaluate model.

```
[17]: # Make predictions
Y_pred = model.predict(X_test)

# Evaluate the model
print("\nClassification Report:")
print(classification_report(Y_test, Y_pred))

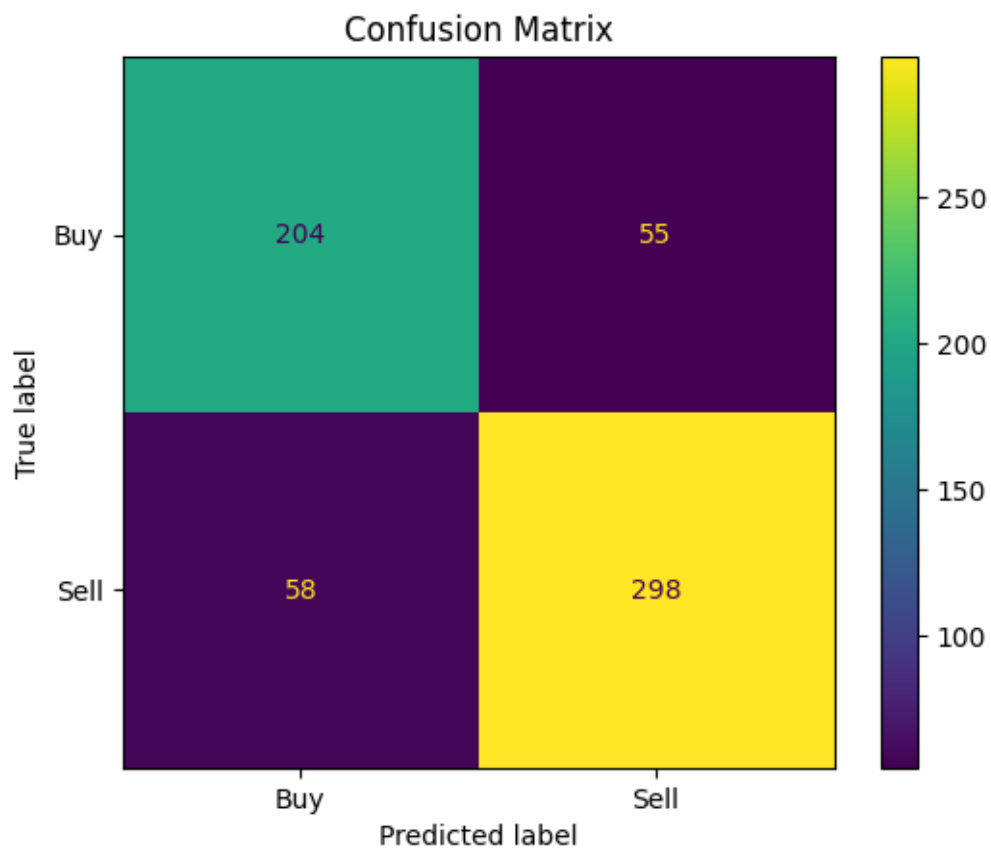
# Confusion Matrix
ConfusionMatrixDisplay.from_estimator(model, X_test, Y_test,
↳ display_labels=['Buy', 'Sell'])
plt.title("Confusion Matrix")
plt.show()

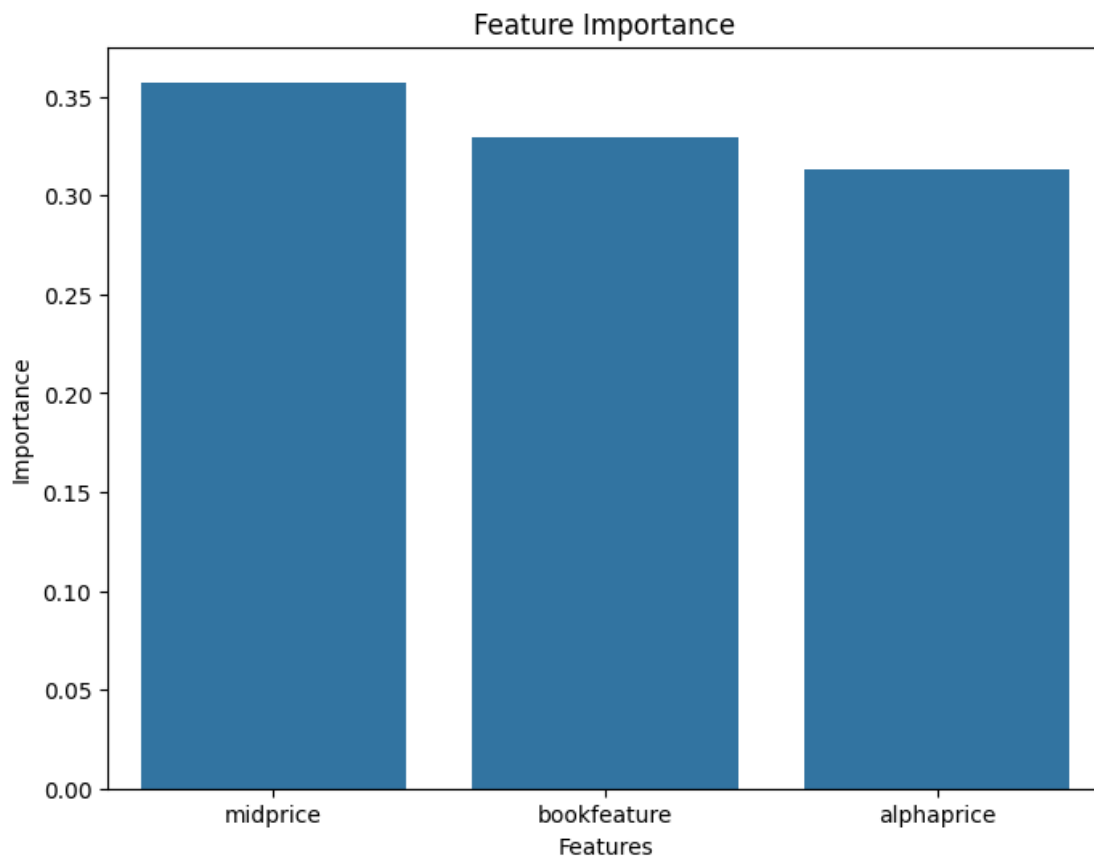
# Feature Importance
importance = model.feature_importances_
plt.figure(figsize=(8, 6))
sns.barplot(x=['midprice', 'bookfeature', 'alphaprice'], y=importance)
plt.title('Feature Importance')
plt.ylabel('Importance')
```

```
plt.xlabel('Features')  
plt.show()
```

Classification Report:

	precision	recall	f1-score	support
0	0.78	0.79	0.78	259
1	0.84	0.84	0.84	356
accuracy			0.82	615
macro avg	0.81	0.81	0.81	615
weighted avg	0.82	0.82	0.82	615





1.4 Task 4

T4-1: Show the output images and texts of running the program.

Unfortunately, I was not able to run it using my GPU, but here is output using CPU:

Model: "model"

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, 28, 28, 1)]	0
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 32)	9248
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 32)	0

```
conv2d_transpose (Conv2DTr  (None, 14, 14, 32)          9248
anspose)

conv2d_transpose_1 (Conv2D  (None, 28, 28, 32)          9248
Transpose)

conv2d_2 (Conv2D)           (None, 28, 28, 1)          289
```

```
=====
Total params: 28353 (110.75 KB)
Trainable params: 28353 (110.75 KB)
Non-trainable params: 0 (0.00 Byte)
-----
```

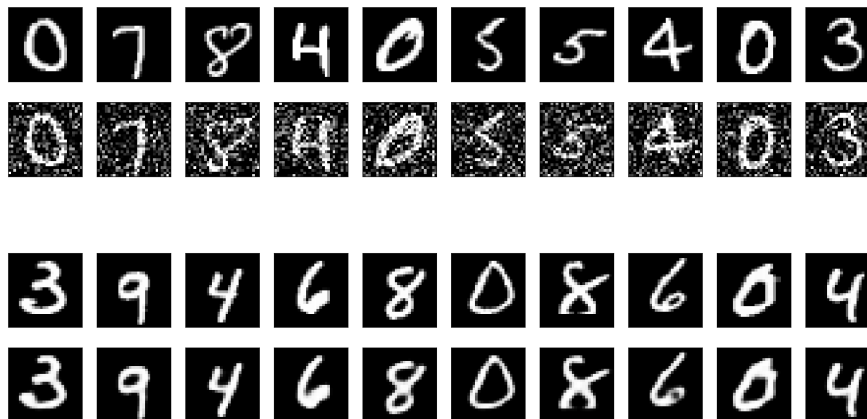
Epoch 1/2

469/469 [=====] - 51s 108ms/step - loss: 0.1300 - val_loss: 0.0732

Epoch 2/2

469/469 [=====] - 46s 99ms/step - loss: 0.0716 - val_loss: 0.0695

313/313 [=====] - 3s 10ms/step



T4-2: How long does it take to run in your environment?

Around 2 minutes.

T4-3: What do you think line 50 and 51 do? Overall, what do you think this program is doing? Explain briefly in your own words (Don't explain the code). Of course, you can search in the Internet!

Lines 50 and 51 are running preprocessing functions for both train and test datasets by normalizing pixel values (scaling them to a range between 0 and 1) and reshaping the data to add a channel dimension, making it compatible with the neural network input format.

In general, this program creates a denoising autoencoder model that learns to remove noise from images of digits by training on noisy data and aiming to reproduce cleaner versions of the original images.